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Your grade: 91.16%

Your latest: **91.16%** • Your highest: **91.16%** • To pass you need at least 80%. We keep your highest score.

Full Course Assessment

1. What is the primary purpose of the Hadamard gate in a quantum circuit? **1 / 1 point**

- To measure the state of a qubit
- To flip the state of a qubit
- (•) To create superposition in a qubit
- To entangle two qubits

Correct
Correct! The primary purpose of the Hadamard gate is to create superposition in a qubit, putting it into an equal superposition of 0 and 1.

2. Which of the following best describes the process of creating entanglement between two qubits? **1 / 1 point**

- Using a series of Pauli-X and Pauli-Z gates on both qubits.
- Applying two Hadamard gates in sequence to both qubits.
- (•) Applying a Hadamard gate to the first qubit followed by a CNOT gate with the first qubit as control and the second qubit as target.
- Applying a Toffoli gate with the first qubit as control qubit and the second qubit as target qubit.

Correct
Correct! Entanglement between two qubits is typically created by first applying a Hadamard gate to the first qubit and then a CNOT gate with the first qubit as control and the second qubit as target.

3. Which quantum gate is primarily used to create superposition in a qubit? **1 / 1 point**

- Pauli-X gate
- CNOT gate
- Toffoli gate
- (•) Hadamard gate

Correct
Correct! The Hadamard gate is used to create superposition in a qubit, transforming it into an equal superposition of 0 and 1.

4. Which of the following are essential steps in running a quantum circuit on the IBMQ Lima quantum computer? **0.8 / 1 point**

- [ ] Save the API token in a file for authentication.
- [x] Connect to the IBMQ backend using Qiskit.

Correct
Correct! Connecting to the IBMQ backend using Qiskit is essential for running the quantum circuit.

- [ ] Implement classical gates in the quantum circuit.
- [ ] Run the circuit on a classical simulator first.
- [x] Obtain an API token from IBM Quantum Experience.

Correct
Correct! Obtaining an API token is a necessary step to access IBMQ Lima.

You didn't select all the correct answers

5. Select the correct statements about the Toffoli gate in quantum computing. **1 / 1 point**

- [ ] The Toffoli gate operates on a single qubit.
- [ ] The Toffoli gate can create superposition in qubits.
- [x] The Toffoli gate is also known as the CCNOT gate.

Correct
Correct! The Toffoli gate is indeed also called the CCNOT gate.

- [x] The Toffoli gate can be used to implement classical logic functions in quantum circuits.

Correct
Correct! The Toffoli gate can implement classical logic functions like AND and OR.

- [ ] The Toffoli gate is a non-reversible gate.

6. Which of the following gates is used to create entanglement between qubits in a quantum circuit? **1 / 1 point**

- S and T gates
- Hadamard and Z gates
- (•) Hadamard and CNOT gates
- X and Y gates

Correct
Correct! The Hadamard gate creates superposition, and the CNOT gate entangles the qubits.

7. What concept does superdense coding demonstrate in quantum computing? **0 / 1 point**

- Measuring a qubit's state without disturbing it
- Transferring two classical bits of information using one qubit
- Creating entanglement between multiple qubits
- (•) Transferring a quantum state from one qubit to another

Incorrect
Incorrect. Superdense coding transfers classical information, not quantum states. Review the main points of superdense coding.

8. Which of the following elements are essential for implementing quantum teleportation in Qiskit? **1 / 1 point**

- [ ] Quantum Fourier Transform
- [x] A pair of entangled qubits

Correct
Correct! A pair of entangled qubits is essential for quantum teleportation.

- [x] CNOT gate

Correct
Correct! The CNOT gate is a key component in the quantum teleportation circuit.

- [ ] An auxiliary classical bit
- [x] Hadamard gate

Correct
Correct! The Hadamard gate is used to create superposition, which is crucial in quantum teleportation.

9. Which gate is used to flip the phase of a qubit? **1 / 1 point**

- XGate
- HGate
- CNOT Gate
- (•) ZGate

Correct
Correct! The ZGate flips the phase of a qubit.

10. What is the role of the ZGate in quantum computing? **1 / 1 point**

- To measure the qubit's state
- (•) To flip the phase of a qubit by 180 degrees
- To transfer a quantum state from one qubit to another
- To create entanglement between two qubits

Correct
Great job! The ZGate flips the phase of a qubit by 180 degrees, which is essential in various quantum algorithms.

11. What is the primary advantage of the Bernstein-Vazirani algorithm over classical algorithms? **1 / 1 point**

- It can solve NP-complete problems.
- (•) It finds the secret number with a single query.
- It does not require any entanglement.
- It uses the Quantum Fourier Transform.

Correct
Correct! The Bernstein-Vazirani algorithm can find the secret number with a single query, demonstrating quantum efficiency.

12. Which of the following are true about the categorization of quantum algorithms based on their advantages over classical computers? Select all that apply. **0.8333333333333334 / 1 point**

- [x] Grover's algorithm can search an unsorted database quadratically faster than any classical algorithm.

Correct
Correct! Grover's algorithm is another example of a quantum algorithm that provides a quadratic speedup for searching unsorted databases.

- [x] Quantum algorithms like Shor's algorithm can factor large numbers more efficiently than classical algorithms.

Correct
Correct! Shor's algorithm is a well-known example of a quantum algorithm that can factor large numbers efficiently, showcasing quantum advantage.

- [ ] Quantum algorithms can solve certain problems exponentially faster than classical algorithms.
- [ ] Quantum algorithms do not offer any advantage over classical algorithms in solving NP-complete problems.
- [ ] Quantum algorithms are always deterministic, providing the same result in every run.
- [ ] All quantum algorithms outperform classical algorithms in every aspect.

You didn't select all the correct answers

13. What does the Hadamard gate do when applied to a qubit in the context of the Deutsch algorithm? **1 / 1 point**

- It flips the state of the qubit.
- (•) It creates an equal superposition of states $|0\rangle$ and $|1\rangle$.
- It applies a phase shift to the qubit's state.
- It measures the state of the qubit.

Correct
Correct! The Hadamard gate creates an equal superposition state which is crucial for the Deutsch algorithm.

14. What steps are involved in executing the Deutsch algorithm on a quantum computer? **1 / 1 point**

- [x] Use the Oracle to apply the function

Correct
Correct! The Oracle is used to encode the function within the quantum circuit.

- [ ] Measure both qubits simultaneously
- [x] Initialize the qubits to

$$|0\rangle$$

and

$$|1\rangle$$

states

Correct
Correct! Initialization of qubits is an important first step.

- [ ] Apply a series of swap gates
- [x] Apply Hadamard gates to both qubits

Correct
Correct! Hadamard gates are applied to prepare the superposition states.

15. Which of the following are applications where Grover's search algorithm can potentially be used? **0.6 / 1 point**

- [ ] Finding the prime factors of a large number
- [ ] Cryptography for breaking symmetric key encryption
- [ ] Solving linear equations
- [ ] Optimizing complex functions in machine learning
- [x] Database search to find a specific entry

Correct
Correct! Grover's algorithm is well-suited for database search problems.

You didn't select all the correct answers

16. What is the primary role of the oracle in Grover's algorithm? **1 / 1 point**

- To entangle all qubits in the circuit
- To perform the Quantum Fourier Transform
- To initialize the qubits in a superposition state
- (•) To mark the correct solution by inverting its amplitude

Correct
Correct! The oracle inverts the amplitude of the correct solution, marking it.

17. Which of the following are essential concepts to understand when learning Shor's algorithm? **1 / 1 point**

- [x] Greatest common divisor

Correct
Correct! The greatest common divisor is a key concept in Shor's algorithm.

- [x] Modular arithmetic

Correct
Correct! Modular arithmetic is essential for understanding Shor's algorithm.

- [x] Periodicity

Correct
Correct! Periodicity is fundamental to Shor's algorithm.

- [ ] Linear regression
- [ ] Graph theory

18. Which of the following are true about the Quantum Fourier Transform (QFT) in the context of Shor's algorithm? **1 / 1 point**

- [ ] QFT is used to initialize the quantum state.
- [ ] QFT must be applied twice in Shor's algorithm to get the correct result.
- [x] QFT helps in converting the periodicity of the quantum state into a measurable form.

Correct
Correct! The QFT is essential for transforming the periodicity into a form that can be measured.

- [ ] QFT is irrelevant if the periodic function is already known.
- [x] QFT is applied after modular exponentiation to find the period of the function.

Correct
Correct! The QFT is applied after modular exponentiation to reveal the period of the function.

19. Which of the following libraries can be used to implement Shor's algorithm? **1 / 1 point**

- [x] Qiskit

Correct
Correct! Qiskit is a widely used library for implementing Shor's algorithm.

- [ ] TensorFlow
- [x] Microsoft's Quantum Development Kit

Correct
Correct! Microsoft's Quantum Development Kit can be used to implement Shor's algorithm.

- [x] Google's Cirq

Correct
Correct! Google's Cirq is another library that can be used to implement Shor's algorithm.

- [ ] Pandas

20. Which gate is used for modular exponentiation in Shor's algorithm? **1 / 1 point**

- Hadamard gate
- (•) Uf gate
- CNOT gate
- Pauli-X gate

Correct
Correct! The Uf gate is used for modular exponentiation in Shor's algorithm.

Coach
Ready to review what you've learned before starting the assignment? I'm here to help.
Help me practice Let's chat

Assignment details
Due: Dec 20, 11:59 PM IST
Attempts: Unlimited
Retry
Submitted: Dec 12, 3:31 AM IST
Your grade: 91.16%
View submission See feedback